Hippolyte Wallaert

+33 642093440 | hippowal@stanford.edu | LinkedIn | GitHub

Education Background

Stanford University 2025 – 2027
M.S. in Computational and Mathematical Engineering (ICME) | GPA: 4.07/4.3 *Stanford, California*
- Key courses : *Safe & Constrained AI*, *Aligning Superintelligence*, *Reinforcement Learning*, *Parallel Computing*, *Numerical Linear Algebra*, *ML with Graphs*.

Ecole Polytechnique 2022 – 2025
Diplôme d'ingénieur with major in Applied Mathematics | GPA: 3.94/4.0 | Top 5 % *Palaiseau, France*
- Key courses : *Optimization & Control*, *Statistics*, *Machine/Deep Learning*, *Advanced Deep Learning*, *Operations research*, *Computer Vision*, *Stochastic calculus*, *Markov Chains & Martingales*, *Advanced Real Analysis*.

Lycée Sainte-Geneviève 2020 – 2022
Preparatory program (Undergrad Maths & Physics) | GPA: 4.0/4.0. | Top 3 % *Versailles, France*

Industry Experience

Linagora June 2024 – Aug. 2024
Software Engineer Intern *Grande Baie, Mauritius*
- Engineered a decentralized Matrix.org messaging server using TypeScript; streamlined API integration.

Research Experience

Research Assistant - Mechanistic Interpretability Apr. 2026 – Present
Supervised by Prof. Éric Darve *Stanford, CA*
- Investigating model steering and nonlinear internal representations to improve fine-grained control of large language models on multiple topics while maintaining coherence and fluency.

Hard-Constrained Neural Networks Jan. 2026 – March. 2026
Supervised by Prof. Madeleine Udell *Stanford, CA*
- Diagnosed gradient filtering in differentiable repair layers and developed an augmented-Lagrangian training approach to prevent vanishing or exploding gradients under convex and nonlinear constraints. [Poster] presented at ICME Research Symposium.

Research Assistant - Priorities and Preferences in Organ Allocation Dec. 2025 – June 2026
Supervised by Prof. Itai Ashlagi *Stanford, CA*
- Analyze kidney transplantation data to study how allocation priorities and patient preferences shape organ-allocation outcomes.

Visiting Student Researcher Apr. 2025 – Aug. 2025
Massachusetts Institute of Technology, supervised by Prof. Alexandre Jacquillat. *Cambridge, MA*
- Developed an exact algorithm for two-stage linear optimization with black-box recourse, constructing Benders-type cuts from zero-order oracle evaluations.
- Derived algebraic tests that bound oracle queries independently of the first-stage dimension; validated the method on facility-location instances motivated by robotic warehousing. [Report]

Selected Projects

Physics-Informed Graph Machine Learning Oct. 2025 – Dec. 2025
- Developed HeGGA (Heterogeneous Graph with Global Attention) for AC-Optimal Power Flow and enforced Kirchhoff's laws through a physics informed loss to improve the accuracy-feasibility Pareto frontier. [Article]

Autonomous Trajectory Planning Sept. 2023 – June 2024
Supervised by Prof. Éric Goubault *Palaiseau, France*
- Optimized stratospheric balloon navigation through non-linear wind fields; benchmarked Deep Reinforcement Learning agents against (1+2)D ResNet architectures for interpretability and reliability.

Skills

Language: English (fluent), French (native), Chinese (basic)

Programming: Python (PyTorch, scikit-learn, pandas), Julia, JavaScript/Typescript, Cuda/C++, Gurobi, Java

Leadership and Activities

French Military Golf Team | National & CISM World Gold Medalist 2024 – 2025
XForum (Career Fair) | Chief Organizer of Universities Fair (1,000+ students) 2024
French Air Force Sept. 2022 – Mar 2023
Assistant to Company Commander (Military Service) *Saint-Dizier, France*
- Led surveillance missions and trained new recruits at Saint-Dizier Air Force Base.